

# Interferometer Measurement of Polarization-Mode Dispersion Statistics

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**Abstract**—A new relationship between polarimeter and interferometer measurements of polarization mode dispersion is proposed. This model bridges the gap between interferometer patterns and probability density function of the principal states of polarization group delay. Experimental results supporting the theory are presented.

**Index Terms**—Interferometry, optical fiber measurements, optical fiber polarization, polarimetry.

## I. INTRODUCTION

WITH the recent development of very high-speed optical networks, the subject of polarization-mode dispersion (PMD) has become more and more important. Recently, the attention of researchers has focused mainly on PMD emulation and compensation, which are crucial subjects for the upgrade of old fiber-optics plants with the new 10- and 40-Gb/s transmission systems. At the same time, the topic of PMD measurements has seen remarkable improvements in the instrumentation technology with the development of high-performance compact instruments for field applications. However, only small contributions have been presented on the measurement theory, assuming that all the basic theoretical work had already been done.

In this paper, we would like to focus our attention on a topic that is usually considered well established: the interferometer method (IM) for PMD measurements. IM is one of the most widely adopted techniques for PMD measurements, especially in the field. We think that, despite its common use, the interpretation of IM suffers from two major weaknesses. The first one is that the theory of light propagation in a randomly birefringent optical fiber, upon which IM is based, does not seem completely satisfactory [1]. The second weakness of IM is the debated question of its relationship with the other measurement techniques, in particular the Jones matrix eigenanalysis (JME) method, which has been addressed in various papers [2], [3] without reaching, in our opinion, an ultimate conclusion.

The purpose of this paper is twofold. First, after a critical analysis of the current theoretical model of IM, we propose a new model for the interferometer measurement of PMD, based on the theory of the principal states of polarization (PSPs). Second, on the basis of this new model, we will show that the interferometer patterns measured by IM are not Gaussian

shaped but approximated measurements of the differential group delay (DGD) probability density function (pdf), as measured by JME. This fundamental result bridges the gap between IM and JME.

This paper is organized as follows. Section II will analyze the IM. The new model for IM analysis is presented in Section III: it will be applied to cases of both deterministic and random PMD. Experimental support to the proposed model will be given in Section IV, comparing results for IM and JME and commenting on the supposedly Gaussian behavior of IM interferometer patterns, so crucial to the interpretation given in [1].

## II. ANALYSIS OF THE INTERFEROMETER METHOD

The interferometer measurement technique, as described in [1], is based upon a well-known theoretical model of propagation in a randomly birefringent optical fiber: the so-called waveplate model. This model consists of cascading a large number of randomly oriented birefringent waveplates, representing the local birefringence random fluctuations of a real optical fiber. This model is used in [1] to perform the analytical calculation of the signal time-of-flight distribution, assuming that, at each interface between a waveplate and the subsequent one, the optical field is randomly coupled to slow and fast polarization axes of the second waveplate. The final result is a complicated analytic expression for the time-of-flight distribution that tends to a Gaussian-like distribution when the fiber is much longer than the polarization coupling length. However, it should be observed that there are no evident physical reasons to identify a time-of-flight distribution calculated in this way (without any reference to the coherence and bandwidth of the optical field) with an electric field distribution, as has been done in [1].

The remainder of [1] is dedicated to the description of an interferometer setup for PMD measurements similar to that shown in Fig. 1. The light source is a broad-band incoherent one followed by a polarizer, and a Michelson interferometer is used to analyze the polarization-mode delays at the exit of the fiber under test. The envelope of the interference fringes originated when scanning the time delay of the Michelson interferometer [i.e.,  $T = 2(n/c)(L_1 - L_2)$ ] is proportional to the autocorrelation of the electric field entering the interferometer itself. Since [1] assumes a Gaussian field distribution at the output of a fiber much longer than the polarization coupling length, the measured autocorrelation function should be Gaussian too, with a standard deviation twice as large as the standard deviation of the field distribution. In our opinion, this conclusion is questionable because it is based on the identification of the signal time-of-flight distribution with the electric field distribution.

Manuscript received May 16, 2001; revised April 4, 2002.

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Digital Object Identifier 10.1109/JLT.2002.800341

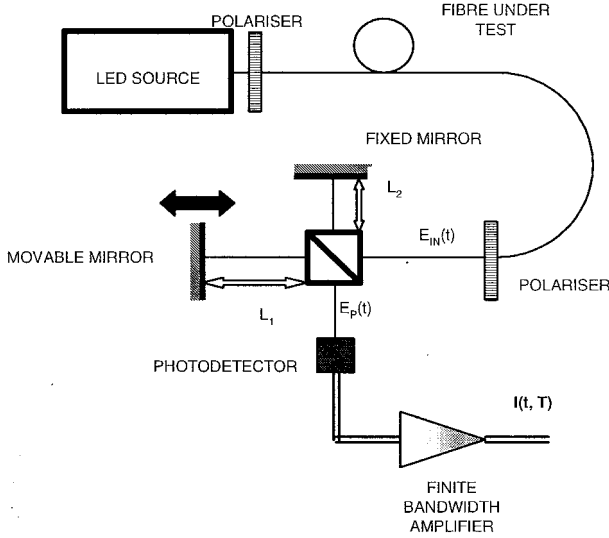


Fig. 1. Setup for interferometer PMD measurements.

A proper analysis of the IM method should include suitable modeling of the optical source, of the light propagation in a fiber with PMD, and of the interferometer detection scheme. These three points will be the subjects of the next three sections.

### III. A PHYSICAL MODEL OF INTERFEROMETER MEASUREMENTS

In this paper, the calculation of the interferometer patterns is performed under the assumption of broad-band radiation, whose spectral components propagate consistently with the PSP theory. With reference to the setup for IM measurements, plotted in Fig. 1, the mathematical procedure is based on the following steps.

- 1) The incoherent source spectrum is staircase-approximated by a sequence of rectangular spectral slots, each described by its amplitude and by its random polarization state and random phase factor [see (1)].
- 2) The slot spectral width is specified with respect to PSP bandwidth and source line width to obtain a sufficiently accurate approximation to the actual DGD.
- 3) A statistical description of light propagation inside the fiber under test is accomplished in terms of the defined slot ensemble that excites a statistical set of PSPs.
- 4) The output signal spectrum is obtained, which depends on four sets of random parameters: projection amplitudes onto PSPs and onto the polarization analyzer axes, DGD values, and phase factors [see (2)].
- 5) The interferometer photocurrent is expressed through the source power spectrum [see (3)].

The outcome of step 5) actually constitutes the main result of this paper: it will be applied to the interpretation of IM and JME experimental results.

#### A. The Spectral Source Staircase Approximation

According to the model, illustrated in Fig. 2, the field spectrum  $S_{in}(\nu)$  of a randomly polarized source is sliced into  $M-N$

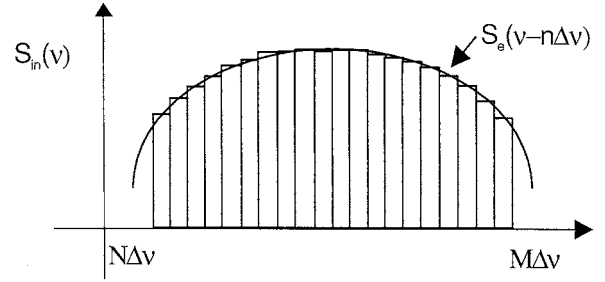


Fig. 2. Representation of the source field spectrum as the sum of elementary slots.

rectangular spectral slots  $S_e(\nu)$  with extent  $\Delta\nu$  and amplitude  $U_n$

$$S_{in}(\nu) = \sum_{n=N}^M U_n \mathbf{p}_n S_e(\nu - n\Delta\nu) e^{j\phi_n}. \quad (1)$$

Each frequency-shifted replica of the fundamental slot has a state of polarization and a phase factor represented by independent random vectors  $\mathbf{p}_n$  and stochastic variables  $\phi_n$ .<sup>1, 2</sup>

The choice of the slot width  $\Delta\nu$  is restricted by two physical bounds. The *upper limit* for  $\Delta\nu$  is the PSP minimum bandwidth in the wavelength range of the source, because each slot is supposed to propagate according to a first-order PSP model. This upper limit is of the order of the reciprocal average DGD for the fiber under test [4] and falls in the gigahertz range even for DGD values of tens of picoseconds. On the other hand, the *lower limit* for  $\Delta\nu$  is set by the electrical amplifier bandwidth of the interferometer, as detailed in Section III-B, and may fit in the 100-kHz range. A value for  $\Delta\nu$  chosen within the mentioned frequency range is considered suitable for an accurate staircase approximation of all light-emitting diodes (LEDs) commonly used in IM setups: this fact confirms that the proposed approximation (1) is physically sound.<sup>3</sup>

Once launched into a fiber, the ensemble  $\{U_n S_e(\nu - n\Delta\nu)\}$  is assumed to excite a set of PSP pairs, each of them characterized by the DGD value  $\Delta\tau_n$ . At the fiber input end, the elementary slots are projected onto the fast and the slow PSPs with amplitudes  $\sqrt{\alpha_n}$  and  $\sqrt{1 - \alpha_n}$ , respectively;  $\alpha_n$  is the power coupling coefficient between the  $n$ th slot and the associated fast PSP axis.

In a similar way, at the fiber output end, we denote by  $\sqrt{\beta_n}$  and  $\sqrt{1 - \beta_n}$  the projection coefficients of the output PSPs onto

<sup>1</sup>Should a (partially) coherent source be used, a (partial) correlation among such variables ought to be introduced accordingly to the emitter actual properties.

<sup>2</sup>Note that if the limit of infinitely narrow frequency slots would be considered, the random sequences  $\{\mathbf{p}_n\}$  and  $\{\phi_n\}$  would be turned into stochastic processes of the optical frequency. On the other hand, if a sufficiently accurate representation of the field spectrum is obtained with slots of small, but finite, spectral extent [compared with step 2) of the work program], this fact should indicate that average random properties could be defined within each slot. In other words, there is no reason to sharpen our frequency “probes” any further: events within each spectral slot can then be characterized by (average) random parameters  $\{\mathbf{p}_n\}$  and  $\{\phi_n\}$ .

<sup>3</sup>At this stage, no explicit choice needs to be supplied for the quantity  $\Delta\nu$  within the operative interval determined just above, since both theoretical predictions and experimental results do not depend explicitly on the actual size of  $\Delta\nu$ . The essential result lies in having shown that an approximation such as that given in (1) actually works.

the interferometer polarization axis. The signal spectrum after this latter polarization selection then reads

$$S_{\text{out}}(\nu) = \sum_{n=N}^M a_n U_n S_e(\nu - n\Delta\nu) e^{-j2\pi\nu\Delta\tau_n/2} e^{j\phi_n} + \sum_{n=N}^M b_n U_n S_e(\nu - n\Delta\nu) e^{j2\pi\nu\Delta\tau_n/2} e^{j\phi_n} \quad (2)$$

where new stochastic variables  $a_n = \sqrt{\alpha_n \beta_n}$  and  $b_n = \sqrt{(1 - \alpha_n)(1 - \beta_n)}$  have been introduced for better convenience. The average DGD of the PSPs has been neglected, referring the  $\Delta\tau_n$ 's at the origin of time delays set by the interferometer.

Since in the time domain the interferometer photocurrent  $I$  is proportional to the autocorrelation function of the output signal, in the frequency domain the relevant quantity is the power spectrum  $|S_{\text{out}}(\nu)|^2$

$$I(\nu) \propto S_{\text{out}}(\nu) \cdot S_{\text{out}}^*(\nu) = \left( \sum_{n=N}^M a_n U_n S_e(\nu - n\Delta\nu) e^{-j2\pi\nu\Delta\tau_n/2} e^{j\phi_n} + \sum_{n=N}^M b_n U_n S_e(\nu - n\Delta\nu) e^{j2\pi\nu\Delta\tau_n/2} e^{j\phi_n} \right) \cdot (\text{c.c.}) \quad (3)$$

where "c.c." means the complex conjugate of the term in parentheses. Next, it will be shown that the interference of different spectral slices  $S_e(\nu - n\Delta\nu) e^{j\phi_n}$  does not give any significant contribution to the photocurrent.

### B. Interferometer Measurement Modeling

As observed in Section II, the IM technique for PMD measurements makes use of an interferometer to detect the autocorrelation of the optical field at the fiber output. As a first step in modeling this approach, a detailed analysis of the output photocurrent will be given, with a particular emphasis on the effects due to the electric amplifier and detector finite bandwidth.

The setup is shown in Fig. 1. Since beyond the input polarizer all optical fields have the same linear polarization, the scalar optical field inputting the photodiode is given by

$$E_P(t, T) = \frac{1}{2}[E_{\text{IN}}(t) + E_{\text{IN}}(t + T)]$$

where  $T = 2(n/c)(L_1 - L_2)$  is the time delay due to the interferometer imbalance; see Fig. 1.

The subsequent processes of quadratic detection and electric amplification are shown in the block diagram of Fig. 3, where these two signal elaborations are considered separately. It should be noted that all signals depend upon two time variables  $t$  and  $T$ .  $t$  is the time variable over which signals are measured: its domain of definition is the measurement time interval (typically, a few seconds). The variable  $T$  is, as previously said, the time delay introduced by the interferometer unbalance, with a domain of definition coincident with the DGD interval of interest, say,  $0.1 \div 100$  ps.

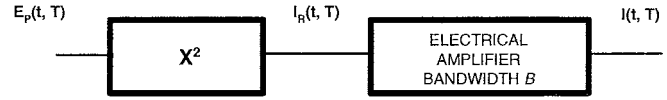


Fig. 3. Block diagram of signal detection.

Omitting for the sake of brevity the detector responsivity  $R = \eta e/(h\nu)$ , the relations among output photocurrent  $I$ , detector photocurrent  $I_R$ , and input field  $E_P$  read

$$I_R(t, T) = E_P^2(t, T) = \frac{1}{4}[E_{\text{IN}}(t) + E_{\text{IN}}(t + T)]^2 = \frac{1}{4}[E_{\text{IN}}^2(t) + E_{\text{IN}}^2(t + T) + 2E_{\text{IN}}(t)E_{\text{IN}}(t + T)]$$

$$I(t, T) = I_R(t, T) * h(t) = \frac{1}{4}[E_{\text{IN}}^2(t) + E_{\text{IN}}^2(t + T) + 2E_{\text{IN}}(t)E_{\text{IN}}(t + T)] * h(t)$$

where  $h(t)$  is the band-limited amplifier impulse response; it obviously depends on variable  $t$  only. As usual, it must be considered that the input fields  $E_{\text{IN}}(t)$  and  $E_{\text{IN}}(t + T)$  oscillate at optical frequencies, the cycles of which cannot be followed by any electrical amplifier. Therefore, the output current should be put in the following form:

$$I(t, T) = I_R * h(t) \propto \langle E_{\text{IN}}^2(t) \rangle_{1/B} + \langle E_{\text{IN}}^2(t + T) \rangle_{1/B} + 2E_{\text{IN}}(t)E_{\text{IN}}(t + T) * h(t)$$

where the symbol  $\langle \cdot \rangle_{1/B}$  stands for a time average calculated over an interval of the order of the reciprocal amplifier bandwidth  $B$ . Of course, the beating term also will have components at electrical frequencies. Since the dc component carries no information and is suppressed by the PMD-meter setup, it is reasonable to confine the attention to the interference contribution only, that is

$$I_{\text{AC}}(t, T) \propto E_{\text{IN}}(t)E_{\text{IN}}(t + T) * h'(t). \quad (4)$$

For a given electrical band of the amplifier, the spectrum of  $h'(t)$  coincides with that of  $h(t)$ , apart from a neighborhood of the origin where  $h'(t)$  vanishes, eliminating dc terms. In the frequency domain, one correspondingly obtains

$$\tilde{I}_{\text{AC}}(f, T) \propto [\tilde{E}_{\text{IN}}(f) * \tilde{E}_{\text{IN}}(f) e^{j2\pi f T}] \cdot \tilde{H}'(f) \quad (5)$$

where Fourier transforms (here indicated explicitly by  $\sim$ ) are taken with respect to the  $t$  variable. Equation (5) immediately shows that spectral components of the convolution term, falling outside the amplifier bandwidth, are cut off by the device. Since the input signal  $\tilde{E}_{\text{IN}}(f)$  is broad band, typically over a range of a few terahertz, a reasonable approximation to (5) could be obtained by expanding the input spectrum in a suitable sum of  $M - N$  spectral slots  $S_e(f - n\Delta f)$ , centered at frequency  $n\Delta f$ ; see Fig. 2

$$\tilde{E}_{\text{IN}}(f) = \sum_{n=N}^M U_n S_e(f - n\Delta f) e^{j\phi_n}. \quad (6)$$

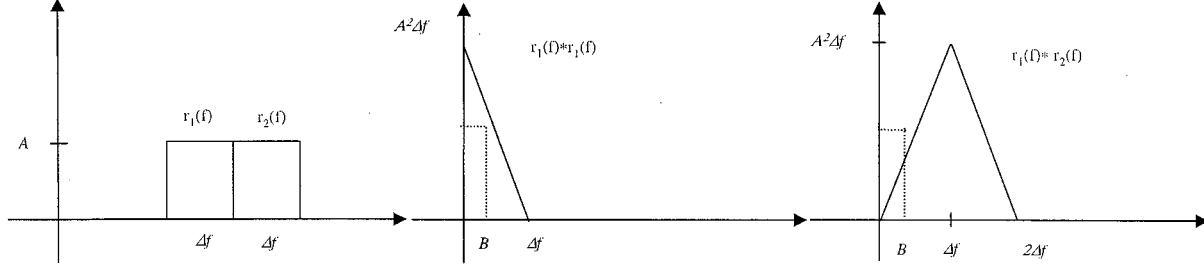


Fig. 4. Representation of two adjacent spectral slots of the convolution of one of them with itself and of the convolution between the two.

Upon substitution of this expansion in (5), one obtains

$$\tilde{I}_{AC}(f, T) \propto \left[ \sum_{n=N}^M U_n S_e(f - n\Delta f) e^{j\phi_n} * \sum_{m=N}^M U_m S_e(f - m\Delta f) e^{j\phi_m} e^{j2\pi f T} \right] \cdot \tilde{H}'(f). \quad (7)$$

We claim that this expression can be reduced to the diagonal form

$$\tilde{I}_{AC}(f, T) \propto \sum_{n=N}^M [U_n^2 S_e(f - n\Delta f) e^{j\phi_n} * S_e(f - n\Delta f) e^{j\phi_n} e^{j2\pi f T}] \cdot \tilde{H}'(f). \quad (8)$$

To prove this statement, consider the convolution of two rectangular spectral slots having amplitude<sup>4</sup>  $A$  and width  $\Delta f$ . It is straightforward to see that, choosing  $\Delta f \geq B$ , only convolutions of one slot with itself or with adjacent slots fall inside  $B$ , as shown in Fig. 4.

For adjacent slots, such a convolution can be calculated as

$$r_1(f) * r_2(f) = \begin{cases} A^2 f, & 0 \leq f \leq \Delta f \\ A^2(2\Delta f - f), & \Delta f \leq f \leq 2\Delta f \\ 0, & \text{elsewhere.} \end{cases}$$

On the other hand, for the same slot

$$r_1(f) * r_1(f) = \begin{cases} A^2(\Delta f - f), & 0 \leq f \leq \Delta f \\ 0, & \text{elsewhere.} \end{cases}$$

To compare the contributions to the photocurrent from adjacent slots and from the same slot, one should compute the ratio between the energy content from the previous two convolutions into the electrical bandwidth

$$R = \frac{E_{11}}{E_{12}} = \frac{\int_0^B [r_1(f) * r_1(f)]^2 df}{\int_0^B [r_1(f) * r_2(f)]^2 df} = 3k^2 - 3k + 1$$

where  $k = \Delta f/B$ . Taking care of the fact that each slot shares two adjacent slots, this ratio has to be halved, resulting in the values represented in Table I. It is apparent that already at  $k = 3$ , the ratio  $R/2$  is around 10 dB and exceeds 20 dB for  $k = 10$ . Therefore, it can be concluded that, in expanding the input spec-

<sup>4</sup>For simplicity, here the same amplitude is assumed for all slots. In view of the source spectrum smoothness, this assumption does not restrict to any extent the validity of the diagonalization in (3).

TABLE I  
THE RATIO  $E_{11}/(2E_{12})$  IN LINEAR AND LOGARITHMIC SCALES FOR DIFFERENT VALUES OF THE PARAMETER  $k = \Delta f/B$

| $\Delta f/B$ | $R/2$ (lin.) | $R/2$ (dB) |
|--------------|--------------|------------|
| 2            | 3.5          | 5.45       |
| 3            | 9.5          | 9.79       |
| 4            | 18.5         | 12.68      |
| 5            | 30.5         | 14.78      |
| 6            | 45.5         | 16.59      |
| 10           | 135.5        | 21.32      |

trum as in (6), when the spectral amplitude  $\Delta f$  of the component slots is greater than three or four times the electrical bandwidth  $B$ , then the photocurrent can be calculated as in (8), i.e., as a sum of diagonal contributions from each slot. This consideration matches the results already discussed in Section III-A for the operative domain of definition of the spectral slot thickness.

### C. Modeling Result

Following the derivation of Section III-B, only diagonal terms in the sum of (3) survive and  $I(\nu)$  takes the form

$$I(\nu) \propto \sum_{n=N}^M (a_n^2 + b_n^2) U_n^2 |S_e(\nu - n\Delta\nu)|^2 + \sum_{n=N}^M (a_n b_n) U_n^2 |S_e(\nu - n\Delta\nu)|^2 \cdot (e^{-j2\pi\nu\Delta\tau_n} + e^{j2\pi\nu\Delta\tau_n}). \quad (9)$$

Equation (9) is the most general expression of the interferometer photocurrent in the frequency domain, which can be obtained without any assumption on the statistical nature of the PMD. In the following two sections, the cases of deterministic and random PMD are separately discussed. It will be shown that the presented model trivially gives the results for deterministic PMD and that it is quite appropriate to cope with the more complex (and, actually, more interesting) case of a random polariza-

tion, disclosing a new way for the interpretation of IM and JME experimental results.

#### D. Deterministic Birefringence

A fiber with a deterministic birefringence (e.g., a polarization-maintaining fiber) is characterized by having the DGD constant and the PSPs unchanging over wavelength. As a consequence,  $\Delta\tau_n = \Delta\tau$ ;  $a_n = a$  and  $b_n = b$  are fixed numbers; and (9) simplifies to

$$I(\nu) \propto (a^2 + b^2) \sum_{n=N}^M U_n^2 |S_e(\nu - n\Delta\nu)|^2 + ab(e^{-j2\pi\nu\Delta\tau} + e^{j2\pi\nu\Delta\tau}) \sum_{n=N}^M U_n^2 |S_e(\nu - n\Delta\nu)|^2$$

that is

$$I(\nu) \propto (a^2 + b^2) |S_{in}(\nu)|^2 + ab(e^{-j2\pi\nu\Delta\tau} + e^{j2\pi\nu\Delta\tau}) |S_{in}(\nu)|^2.$$

An inverse Fourier transform directly gives the following expression for the photocurrent in the time domain

$$I(T) \propto (a^2 + b^2) s_c(T) + ab[s_c(T - \Delta\tau) + s_c(T + \Delta\tau)] \quad (10)$$

where  $s_c(\tau)$  is the source autocorrelation function, corresponding to the inverse Fourier transform of  $|S_{in}(\nu)|^2$  [5]. The photocurrent expression exhibits three components: the source autocorrelation function and its replicas translated around the two symmetrical points  $\pm\Delta\tau$ . This is a simple and well known result [1], which is obtained by straightforward calculations with the proposed model.

#### E. Random Birefringence

The case of fibers with random birefringence is the most important in practice and requires a more careful mathematical treatment. The analysis starts with the calculation of the average of the stochastic process  $I(\nu)$ . This step has a twofold importance. On the one hand, it leads easily to a very elegant formal result, as will be shown in (11). On the other hand, from the measurement point of view, this mathematical operation corresponds to a simple average of independent PMD measurements, as will be shown in Section IV, allowing us to supply an experimental verification of the proposed model. From (9), we calculate the statistical average  $\langle \cdot \rangle$  of the photocurrent

$$\langle I(\nu) \rangle = \left\langle \sum_{n=N}^M (a_n^2 + b_n^2) U_n^2 |S_e(\nu - n\Delta\nu)|^2 + \sum_{n=N}^M a_n b_n U_n^2 |S_e(\nu - n\Delta\nu)|^2 \cdot (e^{-j2\pi\nu\Delta\tau_n} + e^{j2\pi\nu\Delta\tau_n}) \right\rangle.$$

Taking into account the linearity of this average and the fact that the random variables  $\Delta\tau_n$  are independent of  $a_n b_n$  and

( $a_n^2 + b_n^2$ ) (this assumption clearly fails to hold for deterministic PMD, and in fact it has not been introduced in Section III-D), one readily obtains

$$\langle I(\nu) \rangle = \sum_{n=N}^M \langle a_n^2 + b_n^2 \rangle U_n^2 |S_e(\nu - n\Delta\nu)|^2 + \sum_{n=N}^M \langle a_n b_n \rangle U_n^2 |S_e(\nu - n\Delta\nu)|^2 \cdot \langle e^{-j2\pi\nu\Delta\tau_n} + e^{j2\pi\nu\Delta\tau_n} \rangle.$$

Since the statistical average of the exponential of the random variable  $\Delta\tau_n$  is its characteristic function  $\Psi_{\Delta\tau_n}(\nu)$  [5], the previous equation can be cast in the form

$$\langle I(\nu) \rangle = \sum_{n=N}^M \langle a_n^2 + b_n^2 \rangle U_n^2 |S_e(\nu - n\Delta\nu)|^2 + \sum_{n=N}^M \langle a_n b_n \rangle U_n^2 |S_e(\nu - n\Delta\nu)|^2 \cdot (\Psi_{\Delta\tau_n}(-\nu) + \Psi_{\Delta\tau_n}(\nu)).$$

Moreover, for a given fiber, the characteristic function  $\Psi_{\Delta\tau_n}(\nu)$  and the statistical averages  $\langle a_n b_n \rangle$  and  $\langle a_n^2 + b_n^2 \rangle$  do not depend on wavelength; i.e., they do not depend on the sum index  $n$ . Hence, the previous equation simplifies to

$$\langle I(\nu) \rangle = \langle a^2 + b^2 \rangle \sum_{n=N}^M U_n^2 |S_e(\nu - n\Delta\nu)|^2 + \langle ab \rangle (\Psi_{\Delta\tau}(-\nu) + \Psi_{\Delta\tau}(\nu)) \sum_{n=N}^M U_n^2 |S_e(\nu - n\Delta\nu)|^2.$$

The inverse Fourier transform of the signal leads to the statistical average of the photocurrent in the time domain [5]

$$\langle I(T) \rangle = \langle a^2 + b^2 \rangle s_c(T) + \langle ab \rangle s_c(T) * [f_{\Delta\tau}(-T) + f_{\Delta\tau}(T)] \quad (11)$$

where  $f_{\Delta\tau}(T)$  is the pdf of  $\Delta\tau$  and the asterisk indicates the convolution operation.

Also in this case, the photocurrent is given by three contributions. The first term in (11) is related to the (low) coherence properties of the source and is strongly peaked at the origin. The second and third terms correspond to two symmetrically placed side-lobes, conveying all the information required for the PMD measurement. The resulting average photocurrent expression, despite its symmetry around a peak at  $T = 0$ , is very far from being Gaussian shaped because of sidelobes corresponding to the convolution of  $s_c(T)$  with the differential group delay pdf  $f_{\Delta\tau}(T)$ . Hence, as the coherence time of the source has been assumed much shorter than the average DGD, these twin features are approximated replicas of the associated pdf. The quality of this approximation (source-dependent through the contribution of  $s_c$ ) will be assessed on experimental grounds and could be improved by a proper deconvolution of the source autocorrelation function  $s_c$  itself. Moreover, even if the assumption of the independence of variables  $\Delta\tau_n$  from  $a_n b_n$  and ( $a_n^2 + b_n^2$ )

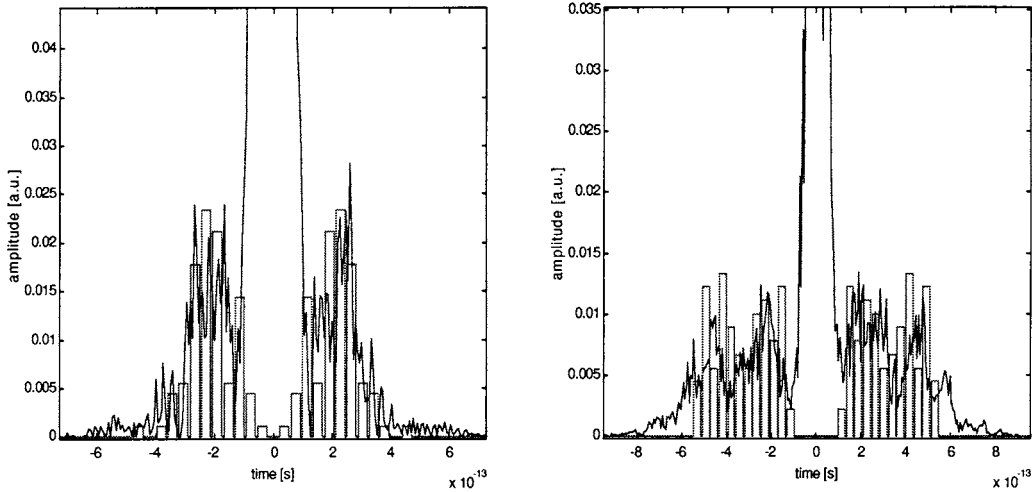


Fig. 5. Interferometer patterns, measured at different temperatures, are shown by a continuous line. The superimposed histograms correspond to the pdf of the DGD, calculated from the JME measurements at the same temperature.

would not hold, the validity of (11) is not limited to the case of random DGD, but formally holds even for polarization maintaining fibers,<sup>5</sup> where  $f_{\Delta\tau}(T)$  is a  $\delta$ -distribution. Under this point of view, (10) can be considered as a special case of (11).

#### IV. EXPERIMENTAL RESULTS

The proposed model can be verified by simultaneous PMD measurements through IM and JME technique. However, as the inherent comparison is based on the independent evaluation of the pdf for the DGD by the two methods, special care must be taken to operate on a large number of independent samples with both experimental procedures.

To this purpose, 50 km of conventional single-mode fiber wound on a drum was placed in a climatic chamber. The temperature was varied from  $\sim 2$  °C to  $\sim 60$  °C through steps of 5 °C.<sup>6</sup> After each thermal step, the fiber temperature was allowed to stabilize, and then PMD was measured in either way. For IM measurements, an LED source having a 3-dB bandwidth ranging from 1480 to 1570 nm was used. JME measurements were performed across the same spectral range, with a wavelength step of 1 nm. Measurements were repeated during a few days, slightly changing at each measurement the starting temperature of the climatic chamber in order to collect a larger number of data. Fifteen independent interferometer patterns and 15 independent JME measurements (with 91 DGD values each) were obtained.

The data were analyzed as follows.

- 1) Each interferometer pattern was normalized.
- 2) The average of the normalized results was taken.

<sup>5</sup>Indeed, even if we cannot speak of independence between the mentioned variables in the deterministic case, it should be observed also that these variables would be no more stochastic in nature, and so they can be manipulated in order to produce the factorization obtained in (11).

<sup>6</sup>This choice allowed us to cover the wider temperature range with the available climatic chamber. It should be stressed that there is no physical reason for which temperatures outside this range should alter the PMD statistics.

- 3) Two symmetric replicas of the DGD probability density function, measured by the JME method, were computed.
- 4) The replicas were compared with the average of the normalized IM patterns.

Two interferometer patterns, measured at different temperatures, are shown in Fig. 5, showing strong fluctuations affecting these measurements. The same figure also includes two histograms representing symmetrical replicas of the DGD probability density function calculated from JME data. Neglecting the convolution with the central peak, which is much narrower than the time distribution of the two side-lobes, a remarkable agreement between IM patterns and JME DGD histograms is found, even if the quality of the pdf reconstruction is limited to a set of  $s = 91$  DGD samples. It should be observed that  $s$  cannot significantly exceed 100 samples if PMD variations of the fiber under test, during the measurement time, are to be neglected, because the time needed for a long series of JME measurements (with  $s \gg 100$ ) implies that the average DGD could substantially change during such characterization.

This is the reason why the normalization and averaging procedure as described in 1) and 2) above is needed for a reliable comparison. The effect of the LED coherence time (around 0.1 ps) is negligible, since it is smaller than the time unit in Figs. 5 and 6.

The averaged interferometer patterns are superimposed to the pdf of the DGD, calculated from the whole set of JME data, in Fig. 6. The agreement is remarkably good, except for the central peak that is not involved in the comparison. The quality of this final superposition highly exceeds that of previous results, shown in Fig. 5, due to the obvious increase of the sample number (in the case of Fig. 6,  $s$  is equivalent to  $15 \times 91$  samples).

Further work will suggest possible approaches to decorrelate the source-dependent effect in (11).

A last comment should be added on the fundamental difference between the model described in this work and that reported in [1]. This is important because of the central role that Gaussian

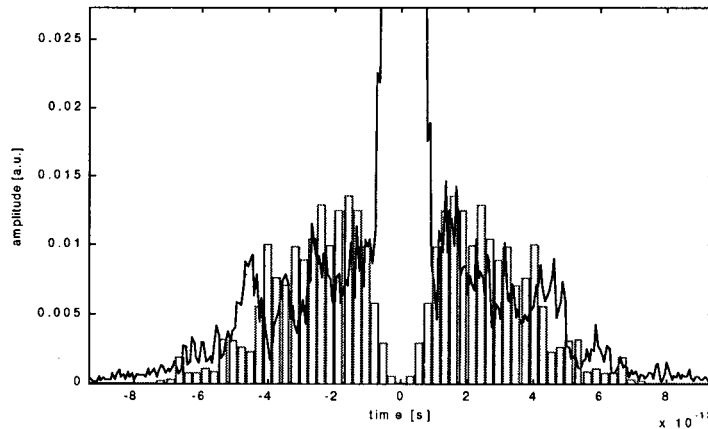


Fig. 6. Comparison of interferometer and polarimeter measurements. The continuous line is the average of 15 normalized interferometer patterns, while the histogram represents two symmetric replicas of the DGD pdf measured from 15 JME measurements.

pdfs play in conjunction with IM previous modeling [1]. As clearly explained in [6], a Gaussian pdf for the DGD of a fiber affected by PMD, can be derived if the propagating radiation is incoherent. This is just the case of the IM method: the condition  $\Delta\tau_s \leq \Delta\tau_{\text{PSP}}$  assures the best visibility of IM interferograms. This consideration governs the operating regime of the interferometer.

However, as the signal detection is performed by an interferometer—which is to be considered in this case a coherent detection technique—different spectral components give independent contributions to the measurement (see Section III-B). As a result, average interferometer patterns contain two symmetric replicas of the group delay pdf, instead of a single Gaussian curve.

## V. CONCLUSION

A new model of the interferometer technique for PMD measurements has been presented and discussed. According to this theory, average interferometer patterns consist of two symmetrical replicas of the DGD pdf (convoluted with the source autocorrelation function). After suitable average, the interferometer patterns match very well the pdf of the DGD measured with the JME technique. This result offers a new physical interpretation of interferometer patterns and clarifies the relationship between interferometer and polarimeter methods of PMD measurement.

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